

SOMUSEKHAR J

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Computer Science undergraduate (CGPA 8.7/10) specializing in applied Machine Learning and AI systems, with hands-on experience building end-to-end platforms spanning NLP, computer vision, and real-time inference. Selected to the Final Round at VIT-AP among 800+ teams nationally for an AI-powered emergency triage platform. Seeking ML/AI engineering roles to apply strong fundamentals in model building and deployment to real-world problems.

EDUCATION

ST. Joseph College of Engineering, Chennai, TN

Bachelor of Engineering in Computer Science [3rd Year]

2023 - 2027

CGPA: 8.7/10.00

SKILLS

Programming: Python, Java, C

ML/DL Frameworks: PyTorch, Scikit-learn, OpenCV, Hugging Face Transformers

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn

Databases: MySQL, PostgreSQL

Cloud & Tools: AWS, Git, GitHub, Google Colab, Jupyter Lab

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Database Management Systems, Operating Systems, Computer Networks

PROJECTS

Indian Sign Language Recognition and Translation | MediaPipe, Random Forest, OpenCV

Feb 2025

- Built a real-time sign-language recognition system using MediaPipe hand landmarks and a Random Forest classifier, achieving **92% classification accuracy** across **26 signs**
- Improved robustness with OpenCV-based real-time feature extraction, cutting latency to **79 ms**

Project Link: <https://github.com/SomusekharJ/Project-SignVisonerZ>

MediRoute AI | Gemini 1.5 Pro, Flask, Python, PyTorch, MySQL, Tesseract OCR

May 2025

- Built an AI-powered emergency triage platform using Gemini 1.5 Pro for symptom reasoning and a custom NLP pipeline for chief-complaint extraction, classifying patient severity into P1-P4 tiers with **90% accuracy**
- Engineered real-time hospital bed/OT availability lookup and GPS-based nearest-facility routing, reducing average routing decision time to **30 seconds**

Automatic Paper Evaluation AI (Theory Papers Collection)

April 2025

- Worked with a team in developing automated descriptive answer grading using OCR and BERT-based NLP in a Flask web app; graded **200+** answer scripts with **90% agreement** to human evaluators

Project Link: <https://github.com/SomusekharJ/EvalVithwaan>

ACHIEVEMENTS

Finalist, VIT-AP V-Launchpad (Funded Project Track)

April 2025

- Selected among 800+ teams from India and 30+ international teams for MediRoute AI, presenting a full pitch deck with financials and go-to-market strategy

Runner-up, Expo 2k26 Project Expo

Feb 2026

- Built a cattle breed classifier in 24 hours using CNN, placing 2nd among 130 teams

CERTIFICATIONS

- Machine Learning Engineer – DataCamp
- Machine Learning in Production – DeepLearning.AI

ADDITIONAL INFORMATION

Languages: English, Tamil, Hindi, and Telugu (Native)

Areas of Interest: Computer Vision, Healthcare AI, NLP, Applied Deep Learning